TECH BIZ SECURITY · SOC ANALYST INTERNSHIP · BATCH B-503

WEEK 1 · TASK 1

Cyber Security Foundations & What Is a SOC?

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01 Objective

This week I set out to build the vocabulary and mental model that every SOC analyst works from. The goals were to understand the CIA triad and how it relates to threats, vulnerabilities and risk; to learn what a Security Operations Center (SOC) does and who works in it; and to see how the defensive blue team and the offensive red team fit together. I completed two TryHackMe rooms as hands-on proof and combined them with the CyberDefenders guide to the SOC analyst role.

02 Proof of Completion

TryHackMe room	Duration	Tasks	Status
Pentesting Fundamentals	30 min	5	Completed, room progress 100%
Security Principles	90 min	9	Completed, room progress 100%

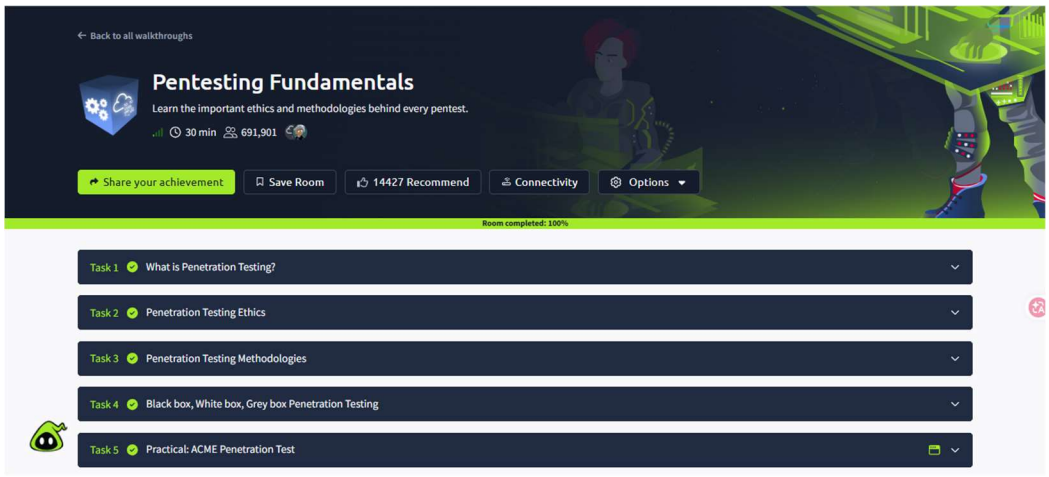


Figure 1. Pentesting Fundamentals: all 5 tasks completed (100%).

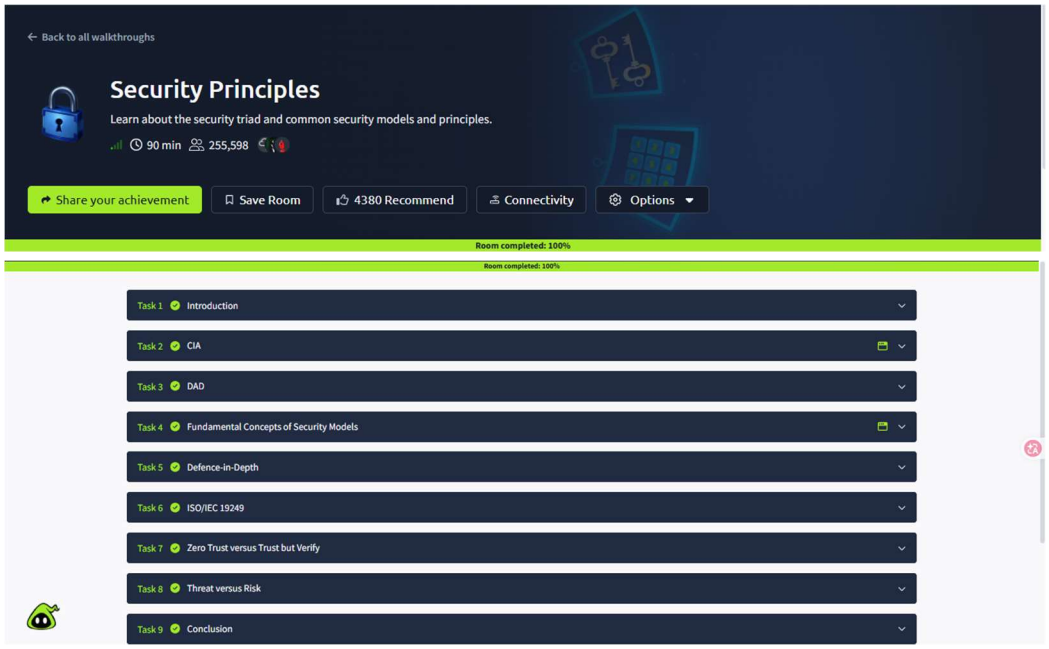


Figure 2. Security Principles: all 9 tasks completed (100%).

03 Security Foundations

3.1 The CIA triad and its opposite, DAD

The CIA triad describes what security is trying to protect. Each property has a mirror image, the outcome an attacker is trying to achieve, which the room calls DAD.

Property	What it means	Attacker's goal (DAD)	Everyday example
Confidentiality	Only the intended people can access the data.	Disclosure	Customer records leaked to the public.
Integrity	Data cannot be altered, and any alteration can be detected.	Alteration	A log file edited to hide an intruder.
Availability	Systems and services are usable when needed.	Destruction / Denial	A DDoS attack takes a website offline.

> **Why this matters in a SOC**
> Almost every alert I will triage can be sorted by the question: which of C, I or A is under attack? It gives me a fast way to judge severity.

3.2 Vulnerability, threat and risk

Term	Definition	Example
Vulnerability	A weakness in a system, process or person.	A server that has not been patched.
Threat	A potential danger linked to that weakness; often an actor who could exploit it.	A ransomware group scanning for that server.
Risk	The likelihood that a threat exploits a vulnerability, combined with the impact on the business.	A high chance of an outage and data loss if the server is hit.

3.3 Security models and principles

- **Bell-LaPadula** protects confidentiality (no reading above your clearance, no writing down to a lower level).
- **Biba** and **Clark-Wilson** protect integrity, so that data is only changed in controlled, trustworthy ways.
- **Defence-in-Depth** uses several layers of protection so that if one fails, another still stands.
- **ISO/IEC 19249** gives five architectural principles (domain separation, layering, encapsulation, redundancy, virtualisation) and five design principles (least privilege, attack-surface minimisation, centralised parameter validation, centralised general security services, preparing for error and exception handling).
- **Zero Trust vs Trust but Verify:** Zero Trust assumes nothing is safe by default and authenticates every request. Trust but Verify grants trust first, then monitors and audits it.

04 Pentesting Fundamentals

A penetration test is an authorised, simulated attack carried out to find weaknesses before a real attacker does. Permission is what separates it from a crime, so ethics come first: written authorisation, a defined scope, agreed rules of engagement and strict confidentiality about what is found.

Methodologies covered: OSSTMM, OWASP, NIST Cybersecurity Framework 1.1 and the NCSC Cyber Assessment Framework (CAF).

Approach	Tester's starting knowledge	Trade-off
Black box	None. The tester acts like an outside attacker.	Most realistic, but slowest and least thorough.
Grey box	Partial, such as a user account or network diagram.	Balances realism and efficiency.
White box	Full access to source code, architecture and credentials.	Most thorough, least like a real attacker.

05 What Is a SOC?

A Security Operations Center is the team, process and technology that continuously monitors an organisation's systems for signs of compromise. Its purpose is to detect threats early, investigate them, and respond before they become a breach. Because attackers do not keep office hours, a SOC works 24/7.

The analyst workflow

Monitor → Triage → Investigate → Escalate or Contain → Document

Roles and tiers

Tier	Role	Responsibilities
L1	Triage analyst	Watches the alert queue, triages alerts, does initial enrichment, and escalates anything real or unclear to L2.
L2	Incident responder	Investigates escalated alerts, scopes incidents, performs deeper forensics, drives containment, and owns confirmed incidents.
L3	Threat hunter / senior analyst	Hunts threats the alerts miss, builds new detections, and handles the hardest forensic and malware work.

Core tools of the trade

Tool	Used for
SIEM	Central log search and correlation; where most triage happens.
EDR / XDR	Endpoint detection, process trees, isolation and response actions.
SOAR	Automated playbooks for repetitive response and enrichment steps.
Threat intelligence platform	Checking IPs, domains and file hashes against known-bad data.
Case management / ticketing	Tracking each alert from open to closed.

Skills to build: log and SIEM analysis, networking and OS fundamentals, endpoint and incident response, threat intelligence frameworks, and scripting.

06 Blue Team vs Red Team

	Blue team (defence)	Red team (offence)
Goal	Prevent, detect and respond to attacks.	Simulate real attackers to expose gaps.
Mindset	Protect and monitor everything, all the time.	Find one way in.
Typical work	Monitoring, log analysis, incident response, hardening.	Reconnaissance, exploitation, privilege escalation, reporting.
Roles	SOC analyst, incident responder, threat hunter.	Penetration tester, red team operator.

> **Where I fit**
> As a SOC analyst I sit on the blue team. Understanding how the red team thinks makes me better at recognising an attack in the logs, so both sides learn from each other.

07 Reflection and Key Takeaways

- Security starts with a clear shared language. Terms such as threat, vulnerability and risk are not interchangeable, and using them precisely will make my reports clearer.
- Integrity matters for my own work too: an investigator who hashes evidence before analysis is applying the I in CIA.
- A SOC is a pipeline, not a single hero. Good triage and clean escalation between L1, L2 and L3 are what make the whole team effective.
- Knowing offensive methods, from black-box to white-box testing, helps me anticipate what an attacker will do next.

Next step: apply these foundations to real log analysis in the coming tasks.

08 References

- TryHackMe: Pentesting Fundamentals (free room)
- TryHackMe: Security Principles (free room)
- CyberDefenders blog: What Is a SOC Analyst?